

1. Rationale for TSDB Selection (InfluxDB)

While the relational PostgreSQL schema ensures strict ACID compliance and avoids data redundancy through normalization, it is fundamentally unsuited for the high-frequency ingestion and querying required by the UCI HAR dataset. In the relational model, sensor readings are stored as double precision [] arrays across normalized tables. Unpacking these arrays for time-based aggregations or windowing introduces significant CPU and I/O bottlenecks.

To address this limitation, we designed an optimized Time-Series Schema using InfluxDB. Time-Series Databases (TSDBs) utilize a column-oriented, time-partitioned storage engine specifically designed to handle high-velocity data streams (50 Hz per sensor in the HAR dataset).

2. Logical Schema Design

Instead of normalized tables with foreign keys, our TSDB schema uses a flattened structure composed of Measurements, Tags (indexed metadata), and Fields (raw numerical values). Below is the conceptual mapping of our TSDB architecture.

Schema Breakdown:

har_kinematic_signals (High-Frequency, 50 Hz):

Stores raw inertial sensor signals. The attention-based anomaly detection framework requires high-resolution temporal data to map attention weights to specific milliseconds of human movement. Storing the 9 axes as separate fields allows the model to ingest a single multivariate record per timestamp.

har_windowed_features (Low-Frequency):

Stores the 561 engineered features (mean, standard deviation, etc.) computed over 2.56-second sliding windows, serving as a baseline for classical machine learning evaluation.

har_kinematic_signals		
TAG	subject_id	1 to 30 - indexed O(1)
TAG	activity	WALKING SITTING STANDING LAYING WALKING_UPSTAIRS WALKING_DOWNSTAIRS
TAG	data_split	train or test
FIELD	tBodyAcc_x	Body acceleration X (g)
FIELD	tBodyAcc_y	Body acceleration Y (g)
FIELD	tBodyAcc_z	Body acceleration Z (g)
FIELD	tBodyGyro_x	Angular velocity X (rad/s)
FIELD	tBodyGyro_y	Angular velocity Y (rad/s)
FIELD	tBodyGyro_z	Angular velocity Z (rad/s)
FIELD	tTotalAcc_x	Total acceleration X (g)
FIELD	tTotalAcc_y	Total acceleration Y (g)
FIELD	tTotalAcc_z	Total acceleration Z (g)
TIMESTAMP	time	UTC ns - 20ms increments (50Hz)

har_windowed_features		
TAG	subject_id	1 to 30 - indexed O(1)
TAG	activity	Activity label
TAG	data_split	train or test
FIELD	f_tBodyAcc_mean_X	Mean body acc X
FIELD	f_tBodyAcc_std_X	Std body acc X
FIELD	f_tBodyGyro_mean_X	Mean gyro X
FIELD	f_fBodyAcc_mean_X	Freq domain mean acc X
FIELD	f_angle_Z_gravMean	Angle Z gravity mean
FIELD	f_556_others	... 556 remaining features
TIMESTAMP	time	UTC ns - 2.56s increments

3. Architectural Advantages for Anomaly Detection

Storage Efficiency:

TSDBs use Delta-of-Delta encoding and Gorilla compression for floating-point values. Storing the 9 raw signal channels in InfluxDB significantly reduces disk usage compared to the array-based PostgreSQL representation.

O(1) Metadata Filtering:

By defining `subject_id`, `activity`, and `data_split` as Tags, the TSDB creates an inverted index. Queries such as retrieving training data for Subject 5 execute in near O(1) time complexity, eliminating costly JOIN operations required in PostgreSQL.

Timestamp Generation Strategy:

Since the UCI HAR dataset uses relative time windows rather than absolute timestamps, our ETL pipeline generates a synthetic continuous timeline (e.g., starting at 2024-01-01T00:00:00Z). For `har_kinematic_signals`, timestamps are incremented by exactly 20 milliseconds per record to reflect the 50 Hz sampling rate.

4. Baseline Benchmarking Protocol (Sprint 1 → Sprint 2 Transition)

To quantitatively demonstrate the performance advantage of the TSDB for time-series aggregation, we define the following benchmarking protocol to be executed in Sprint 2. The test evaluates a standard downsampling operation, aggregating 50 Hz sensor data into 1-second intervals.